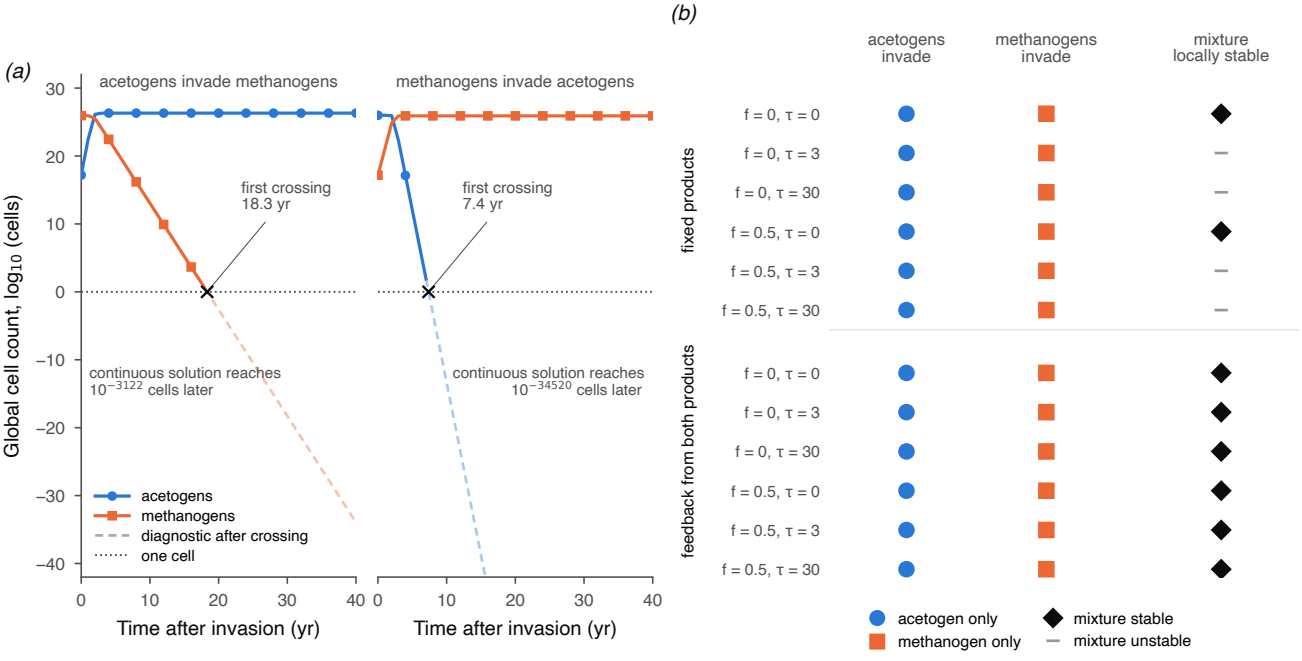
(a)
acetogens invade methanogens
methanogens invade acetogens
Global cell count, log10 (cells)
first crossing
18.3 yr
first crossing
7.4 yr
continuous solution reaches
10−3122 cells later
continuous solution reaches
10−34520 cells later
acetogens
methanogens
diagnostic after crossing
one cell
Time after invasion (yr)
Time after invasion (yr)
(b)
acetogens invade
methanogens invade
mixture locally stable
fixed products
feedback from both products
f = 0, τ = 0
f = 0, τ = 3
f = 0, τ = 30
f = 0.5, τ = 0
f = 0.5, τ = 3
f = 0.5, τ = 30
acetogen only
methanogen only
mixture stable
mixture unstable